Reassessed & Repossessed: Assessment Shocks and Foreclosure

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Abstract

Most U.S. homes are assessed on mandated cycles of three years or longer. Appreciation goes untaxed between assessments and could translate to a discrete increase in the tax liability after the correction. I study 45 countywide assessments in 34 North Carolina counties between 2011 and 2018, comparing each with counties that assess every year by synthetic difference-in-differences, and observe foreclosures for all homes. An assessment raises the quarterly foreclosure filing rate by 17 percent, about 6,700 additional filings over two years across the assessments studied. The additional filings decline steadily with home value, from 4.3 per 1,000 homes over two years in the cheapest quartile to 1.8 in the most expensive. In 18 reassessments where tax liabilities are also observed, each additional percent of the tax increase raises the completed foreclosure rate by 9 percent of its pre-period level. Long assessment cycles concentrate payment shocks, and the foreclosure filings that follow, on the owners least able to absorb them.

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1 Introduction

Most homes in the United States are reassessed on mandated cycles of three years or longer for property taxation.[1] Between reassessments the home's assessed value is fixed, so appreciation over the interval goes untaxed. Then, accumulated appreciation enters the assessed value in a single year, potentially delivering a discrete jump in the owner's tax liability. In contrast, in counties that assess homes annually, property tax liabilities track market values, raising the owner's tax liability in small yearly increments.

Homeowners that are not sufficiently forward looking or are financially strained may experience financial distress after a reassessment. Property tax increases reach mortgaged homeowners as monthly payment changes starting the year after a reassessment, and larger required payments raise the probability of delinquency and foreclosure (Fuster and Willen, 2017; Ganong and Noel, 2020). Escrow, the service mortgage lenders provide that smooths the tax liability over 12 months, already obscures the property tax for owners (Cabral and Hoxby, 2012), and a tax that stays flat for years may give owners little reason to expect a jump. Therefore, owners who fail to account for changes to their property tax liabilities through saving or borrowing may be at risk of default.

On the other hand, homeowners may financially benefit from longer reassessment intervals. Owners subject to longer intervals pay lower tax liabilities than they would under annual assessment if their home's market value rises faster than their jurisdiction's annually adjusted tax rates (Kim and Hou, 2025). An owner of such a home sees tax savings due to this horizontal inequity. If the savings are a sufficient benefit to offset the cost of a sharp upward shift in taxation, longer assessment intervals could be financially beneficial for owners, reducing the risk of default.

Therefore, this paper studies how multi-year assessment cycles affect foreclosure filings and completed foreclosures, where a filing reflects initial default on payments and a completion reflects the loss of the home. To do this, I study 45 countywide assessments in 34 North Carolina counties between 2011 and 2018. North Carolina counties assess on a statutory cycle of four to eight years with no limit on assessment growth, so the date of each assessment is predetermined and the entire accumulated appreciation enters the property tax liability in a single year. I observe recorded foreclosure filings and completions of all owner-occupied homes two years before and after a reassessment. I compare outcomes in these counties to those in 90 counties outside of North Carolina that assess annually via synthetic difference-in-differences (SDID) (Arkhangelsky et al., 2021). SDID weights the comparison counties so that their outcomes move with each treated county's before its assessment, and weights the pre-period quarters that best predict the comparison counties' later path.

[1] The share is computed from the Lincoln Institute's assessment-cycle and assessment-limit tables (2023) and housing stock estimates from the ACS (2024).

I find that long reassessment cycles raise the probability that a home received a filing and completed a foreclosure. I measure each outcome as a quarterly rate per 1,000 owner-occupied homes and average the 45 event estimates with equal weight, which gives the effect of an assessment on the average owner-occupied home. In the two years after a North Carolina reassessment, the quarterly foreclosure filing rate rises by 0.37 per 1,000 owner-occupied homes, 17 percent of its pre-period level of 2.23. This implies about 3.0 additional filings per 1,000 owner-occupied homes over two years, about 100 additional filings in the median treated county of 34,000 homes. Completed foreclosure rates also rise by 17 percent, roughly 1.5 homes per 1,000 over two years, roughly 50 foreclosures for the median county. Applying the average effect across reassessments to all 2.25 million owner-occupied homes covered implies about 6,700 additional foreclosure filings and 3,300 additional completed foreclosures over the two years following assessment.

I test whether the effect of an assessment varies with cycle length, since longer cycles leave appreciation untaxed for more years. I regress each assessment's estimated effect on the number of years since the county's previous assessment. Each additional year is associated with a rise in the quarterly filing and completion rates of two and three percent of their pre-period levels, though neither estimate is distinguishable from zero. I then average the estimates with equal weight within groups of assessments that closed cycles of the same length: 21 four-year, 5 five-year, 4 six-to-seven-year and 15 eight-year cycles. Effects are distinguishable from zero at the 5 percent level only after eight-year cycles, where filings and completed foreclosures rise by 26 and 27 percent of their pre-period rates, about 3.9 and 2.2 per 1,000 homes over two years, consistent with longer cycles leaving more appreciation untaxed.

Scaling each assessment's effect by the tax change it produced is consistent with a tax liability channel for completed foreclosures. For the assessments in which tax liabilities can be measured, I test whether distress scales with the size of the tax change by estimating a Wald ratio (**?**): the average effect of the assessment on each distress outcome divided by its average effect on the tax liability. With the assessment as the single instrument for the tax, this ratio equals the two-stage least squares estimate. A positive ratio would be consistent with distress arising through larger tax liabilities that strain household budgets. Tax liabilities can be measured in 18 of the 45 assessments, in 14 counties, where county rolls record every parcel's assessed value before and after the assessment, so that each home's tax liability can be computed from its assessed value and the published tax rates. Because Florida's statutory rolls permit the same calculation, I estimate the effects of these 18 assessments again against Florida comparison counties alone. I measure the effect on the tax as the change in the mean log tax from the eight quarters before the assessment to the eight quarters after it, in the North Carolina county relative to its weighted Florida comparison.

Over the two years after an assessment, the average tax liability in these North Carolina counties rises by 0.8 to 2.1 percent relative to the Florida comparison, with the range reflecting the separate SDID comparison

weights estimated for each outcome. Dividing the Wald ratio by an outcome's pre-period rate yields an elasticity, the percent change in the distress rate associated with a one percent increase in the tax liability. Each additional percent of tax increase raises the quarterly completed foreclosure rate by 0.11 per 1,000 owner-occupied homes, about 9 percent of its pre-period rate of 1.16. This effect accumulates to about 0.9 completed foreclosures per 1,000 homes over two years and implies an elasticity of roughly 9, with a 95 percent interval of about 1 to 17. The corresponding estimate for foreclosure filings is negative and indistinguishable from zero. These estimates are consistent with larger tax liabilities raising the likelihood that an owner's distress ends in a completed foreclosure, with no detectable effect on the number of owners who enter foreclosure proceedings.

Owners of low value homes may experience more strain due to a higher baseline risk of default and faster home appreciation. Owners of low value homes tend to have low incomes: in the 2019 Survey of Consumer Finances, the median owner in the bottom national quartile of home value earned about $47,000, against $197,000 in the top quartile. Low-income households hold less liquid savings, have lower access to credit, and misperceive tax liabilities more (Jappelli, 1990; Gindelsky et al., 2023). While the average property tax liability can change little in an assessment year due to simultaneous decreases in local tax rates, positive shocks to liabilities fall on homes that appreciated faster than the jurisdiction average.[2] In 26 of the 42 assessments studied here, the fastest appreciating homes were those in the bottom half of the price distribution.[3] Consequently, long assessment cycles may raise distress most for owners of low value homes.[4]

Accordingly, I also study how long assessment cycles affect foreclosure along the price distribution. I estimate each value quartile separately, with its own comparison counties and weights, allowing each quartile's effect to be compared with its own pre-period rate. The rise in foreclosure filings declines steadily with home value: after an assessment, filings in the bottom quartile rise by 20 percent of their baseline against 13 percent at the top. Across all 45 assessments, that is about 2,400 additional filings in the bottom quartile and 1,000 in the most expensive. Completed foreclosures rise by 19 percent of their baseline in the bottom quartile against 13 percent at the top, or about 1,100 and 500 homes over two years across the 45 events. I measure the gap between the average effect of the assessment on the three cheaper quartiles and its effect on the most expensive, whose owners had the lowest foreclosure rates before the assessment. Completed foreclosures for the bottom three quartiles rose by about 15 percent more of their pre-period rate than for the most expensive, translating to 1.4 more homes per 1,000 over two years. Larger increases in distress

[2] Jurisdictions can roll back rates at a reassessment to hold aggregate tax revenue roughly constant.

[3] Calculation from ZTRAX arm's-length sales of owner-occupied homes: homes are sorted into value quartiles by the median sale price of their census block group prior to the assessment window. Appreciation is then measured as the change in a quartile's median sale price from the previous assessment year to the year before the assessment. Median sale prices rose 4 percent from the previous assessment, against 1 percent for the upper half.

[4] Assessment growth caps and homestead exemptions, which disproportionately benefit fast appreciating and low value homes, respectively, do not exist for North Carolina.

among cheaper homes are consistent with their owners being closer to default when the assessment raises their tax liability.

I contribute to research on how property tax assessment affects homeowners and on how required payments affect mortgage default, using a novel dataset. It links each owner-occupied home's annual property tax liability, drawn from county and state tax rolls in North Carolina and Florida, to its recorded deeds and foreclosure filings within each assessment window. Studies of assessment practice document how long intervals between assessments let assessed values drift from market values, and how that drift distributes the tax burden across homes of different value (Berry and Wang, 2024; Kim and Hou, 2025). Studies of mortgage default show that larger required payments raise delinquency and foreclosure, identifying payment effects from interest-rate resets, due-date calendars and program rules (Anderson and Dokko, 2016; Fuster and Willen, 2017; Ganong and Noel, 2020). This paper differs from both. It follows the correction that ends a long assessment interval to its consequences for mortgage distress, the outcome the assessment literature leaves unmeasured. It also uses a payment change whose date is fixed by statute and whose size varies across homes within the same county, which lets the payment response be compared across the value distribution.

2 Background

2.1 County Assessment Practices

A home's property tax liability is the sum of the levies of every local government whose boundaries contain it, typically a county, a city or town, a school district and one or more special districts, each applying its own rate to the home's assessed value, the assessor's estimate of its market value less any exemptions (Amornsiripanitch, 2024; McMillen and Singh, 2020). A home's liability therefore changes when its assessed value changes or when the rates applied to it change.

State law determines how often a county must assess all properties, and a county may choose to assess more often at an administrative cost. Around half of U.S. counties lie in states that require annual assessment, and other states set cycles of three to ten years (Berry and Wang, 2024).[5] Between assessments a home's assessed value is fixed, so it falls further behind the market each year that prices rise, and the gap closes in a single step at the next assessment (Fraenkel, 2024).

Counties set their tax rates every year to meet revenue needs. Many local governments set a target levy and divide it by the tax base, so a rise in assessed values is partly offset by a lower rate (Lutz, 2008). Several states require a jurisdiction to publish the revenue-neutral rate, the rate that would raise its prior revenue on

[5] 1,557 counties of 3,244 total U.S. counties lie in 23 annually assessing states: AL, AK, AZ, FL, GA, HI, ID, KS, KY, MI, MN, NE, NV, NM, ND, OK, OR, SD, TX, UT, WA, WV, WY. Source: Lincoln Institute.

the new values, and to justify any higher rate (Fraenkel, 2024). Where rates adjust this way, an assessment changes total revenue little and mainly reallocates it, raising liabilities for homes that appreciated faster than the county as a whole and lowering them for homes that appreciated more slowly.

State law also limits how much of an assessment reaches the tax liability. Homestead exemptions remove a fixed amount or share of an owner-occupied home's value from the base, and assessment growth caps limit the annual rise in a homestead's taxable value. Where these protections are strong, a large rise in market value reaches the liability gradually, and where they are weak or absent the full correction arrives in the assessment year.

2.2 Assessment in North Carolina

North Carolina requires each county to assess all real property at its market value as of January 1 of the assessment year, at least once every eight years, and a county's commissioners may move an assessment earlier or adopt a shorter cycle by resolution.[6] Counties typically set their schedules several years in advance but can shift them in response to revenue needs or changing markets.

Between assessments a home's assessed value changes only with new construction, demolition or a successful appeal. The state imposes no limit on how much an assessed value may rise and offers no general homestead exemption.[7] Nearly all of an assessment therefore reaches the tax liability of the typical owner-occupied home in the year it takes effect.

In an assessment year, the county mails notices of the new values early in the year and must publish the revenue-neutral rate, adjusted for the average growth of the base through improvements, before adopting its rate, although it may adopt a different one.[8] Tax bills fall due on September 1, so an owner learns the new value months before the first higher tax comes due.

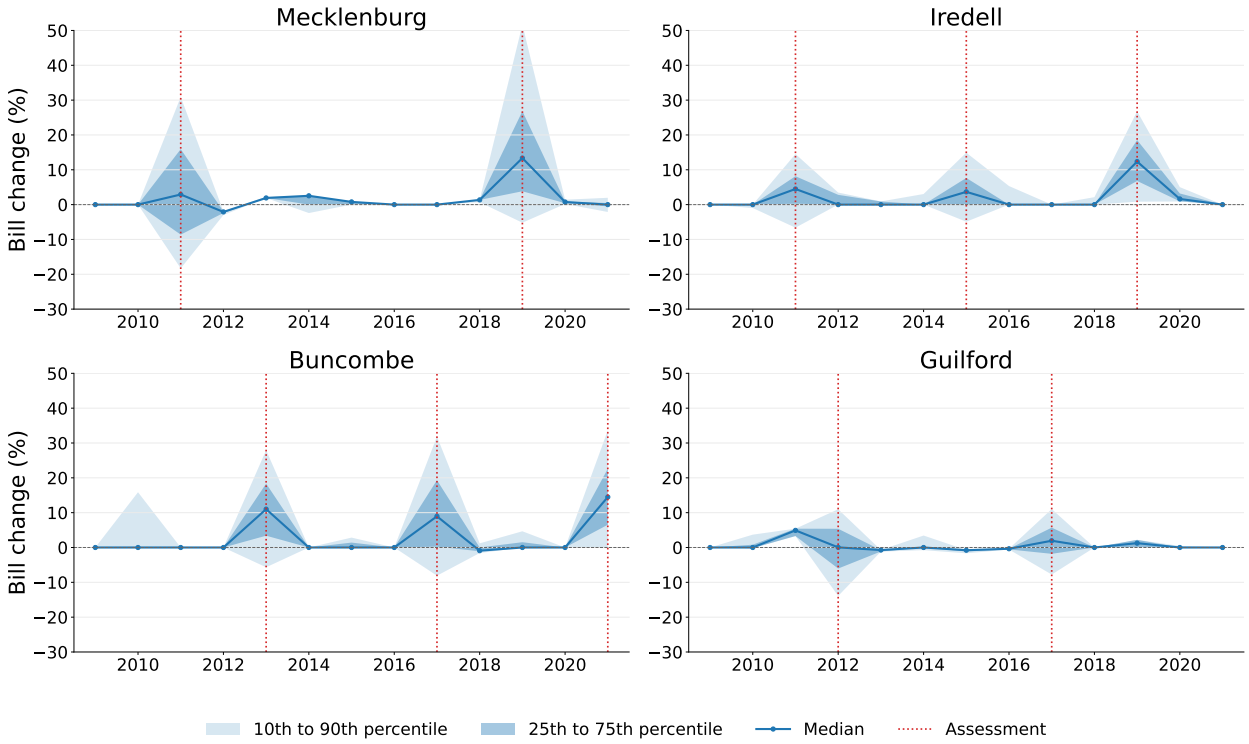
Figure 1 shows the resulting changes in tax liabilities in four counties. In years without an assessment, the middle half of owner-occupied homes see their liabilities change by less than 3 percent, and many by nothing at all; rates can change every year, so revenue grows between assessments through the rate (Appendix Table A2). In an assessment year the distribution widens, and the median liability can rise sharply, by 13 percent in Mecklenburg in 2019. The same assessment raises some liabilities and lowers others. In Mecklenburg in 2011 and Guilford in 2012, the first assessments after the housing crash, the median liability moved by less than 3 percent, while the tenth of homes at the bottom saw liabilities fall by 14 to 18 percent and the tenth at the top saw them rise by 11 to 31 percent.

[6] N.C. Gen. Stat. §§ 105-283, 105-285 and 105-286(a)(3).

[7] Relief is confined to low-income elderly or disabled owners and disabled veterans. N.C. Gen. Stat. §§ 105-277.1, 105-277.1B and 105-277.1C.

[8] N.C. Gen. Stat. §§ 159-11(e) and 159-13.

Figure 1: Annual change in owner-occupied homes' tax liabilities, four North Carolina counties.



Notes. Each home's percent change in its tax liability from the previous tax year. The line is the median and the bands span the 25th to 75th and 10th to 90th percentiles. Dotted lines mark countywide assessments. Mecklenburg and Guilford tax liabilities are recorded on the county rolls; Iredell and Buncombe tax liabilities are computed from assessed values and the combined county, municipal and district rates. Years with fewer than 200 homes billed in both years are omitted.

2.3 Assessment in the comparison counties

The comparison counties lie in nine states and assess every year, so their tax liabilities carry no deferred correction. Some of these states also hold taxable values below the market. Florida and Michigan, which together supply about half of the comparison weight, cap the annual rise in a homestead's taxable value, Florida at 3 percent or the change in consumer prices, whichever is lower, and Michigan at 5 percent or inflation, and Arizona caps it at 5 percent.[9] The counterfactual for a North Carolina assessment is therefore a schedule of small annual changes, and the estimates measure the effect of receiving the correction in one year instead of in smaller intervals.

2.4 How tax changes reach mortgaged owners

Among homeowners in the American Community Survey linked to deed records, 60 percent pay through escrow, 20 percent hold a mortgage without escrow and 20 percent have no mortgage (Gindelsky et al., 2023). An escrow servicer collects a twelfth of the expected annual tax and insurance with each monthly

[9]Fla. Const. art. VII, § 4(d); Mich. Const. art. IX, § 3; Ariz. Const. art. IX, § 18(3)(b).

payment, and federal rules require it to reset that deposit to the projected liability once a year and to spread any shortage of a month's payment or more over at least twelve months.[10] A higher tax therefore raises the monthly mortgage payment within about a year of the assessment, by the increase in the tax plus the repayment of the shortfall the old deposit left. An owner who misses the higher payment is delinquent on the mortgage, and continued delinquency leads to a foreclosure filing and, unless the owner cures the default, sells the home or deeds it to the lender, to a completed foreclosure. A lender can also pay unpaid taxes on a loan without escrow and add the amount to the secured debt, which moves that delinquency onto the mortgage.[11] Owners with no mortgage owe the county alone, so the foreclosure filings and completed foreclosures studied here do not capture their distress.

3 Conceptual Framework

A homeowner could save towards a property tax liability increase from an expected reassessment. Yet, evidence on how households anticipate expenses suggest many do not. Households budget accurately for frequent expenses but underestimate their spending on exceptional expenses, which are also commonly larger (Sussman and Alter, 2012; Karlan et al., 2016). Limited attention to unusual expenses can in turn produce undersaving. Households also fail to smooth changes they could foresee. Spending responds to predictable changes in Social Security tax withholding (Parker, 1999), and it drops sharply at the predictable end of unemployment benefits, a pattern consistent with present-biased or myopic households and inconsistent with rational models of liquidity constraints (Ganong and Noel, 2019). Reassessment at multi-year intervals shares these features. The resulting tax liability change arrives once every four to eight years, and it reaches an escrowed owner through a mortgage payment that already obscures the tax amount (Cabral and Hoxby, 2012). An owner may therefore know that an assessment is scheduled and still be unable to absorb it.

The timing of a required payment can affect default independently of its size. Borrowers whose first lump-sum property tax bill falls within three months of origination have first-year default rates 2 to 6 percent higher than those billed ten to twelve months after origination (Anderson and Dokko, 2016).[12] Mortgage modifications that lower short-term payments by extending maturities have large effects on default, while principal reductions that leave short-term payments unchanged have none (Ganong and Noel, 2020). Assessment cycles determine the timing of tax changes. Under annual assessment the liability changes in small yearly steps, while under a multi-year cycle the accumulated change arrives in one year. Figure 2 shows this for two homes of equal value in a county that raises the same revenue each year: under annual

[10] 12 C.F.R. § 1024.17(c) and (f) (Regulation X).
[11] See, for example, the FHA model mortgage, HUD Handbook 4165.1 REV-2, Appendix II, paragraph 7.
[12] The estimate comes from subprime borrowers.

assessment each tax amount moves by about $43 a year, while under an eight-year cycle each moves by $346 in a single year. Escrow concentrates that change further for borrowers, because the mortgage servicer must raise the monthly deposit to the new liability and also recover the shortfall from the old deposit, usually over twelve months, so the monthly payment first rises by about twice the increase in the tax (Section 2.4). A reassessment should therefore raise mortgage distress relative to annual assessment, even where the county's rate holds its revenue constant, because a bill increase can push a strained owner into default while an equal decrease mostly adds to the cushion of owners already able to pay. Longer cycles leave more appreciation to arrive at once and make the change a rarer event for the household, so the increase in distress should be larger after longer cycles.

Owners of low value homes may be both more likely to misjudge a property tax increase and less able to aborb it. These owners have fewer resources: in the 2019 Survey of Consumer Finances, the median owner in the bottom national quartile of home value earned about $47,000 and held about $2,700 in liquid assets, against about $197,000 and $39,400 in the top quartile. Low-income homeowners also have greater error when judging their own property tax liabilities (Gindelsky et al., 2023). Such gaps can matter for default: among subprime borrowers, weaker numerical ability predicts mortgage default independently of mortgage choice (Gerardi et al., 2013). A misjudged expense therefore leaves an owner of a low value home with less capacity to recover, and increase in tax liabilities after reassessment may raise distress more among low-value housing.

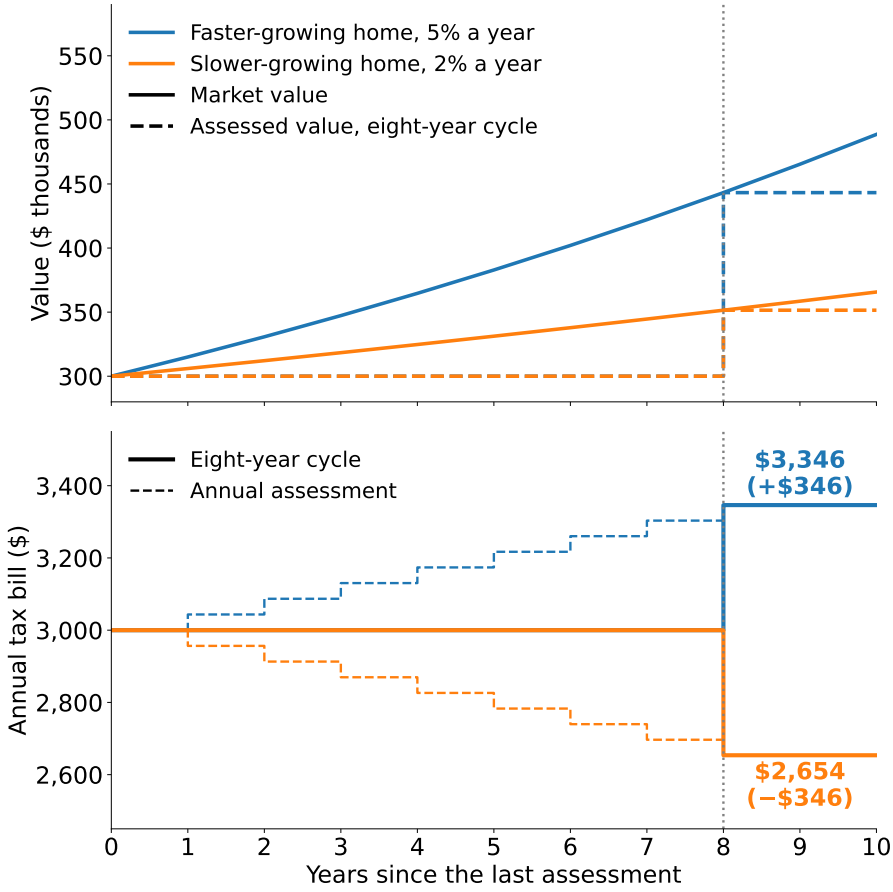


Figure 2: Two homes under annual assessment and under an eight-year cycle.

Notes. Illustration, not estimated. Both homes are worth $300,000 in year 0 and pay $3,000. One appreciates 5 percent a year and the other 2 percent. The county consists of the two homes and sets its rate each year to raise $6,000. Top panel: market values and, under the eight-year cycle, assessed values. Bottom panel: annual liabilities. Shaded areas are the differences between the tax liabilities under the two schedules before the assessment.

An owner whose home appreciated faster than the county's has also paid less tax than under annual assessment, and setting that difference aside would fund the later increase, although a deferred tax is easy to overlook because no statement records a bill the owner did not receive. Even a full saver would only delay the adjustment: the faster-growing home in Figure 2 defers about $1,200 over the cycle, enough to cover its $346 higher bill for about three and a half years, after which the higher bill continues. I do not observe what owners expected or saved, so the estimates cannot separate inattention from an owner who knowingly spent the difference because income and savings were short.

4 Synthetic difference in differences

I estimate the effect of an assessment on foreclosures by comparing each of the 45 North Carolina assessments with counties that assess every year. In those counties assessed values follow market values each year, so appreciation never accumulates untaxed and no discrete correction ever comes due. Their owners face the same regional housing and credit cycles as owners in the treated counties without the jump in tax liabilities. The unit of observation is a county-quarter, and each event is studied over the eight quarters before and after its assessment. The comparison counties stand in for how the treated county's outcomes would have moved had the assessment not occurred.

A pooled regression with county and quarter fixed effects, estimated by ordinary least squares across the 45 assessments, would compare counties assessed at different dates, and eleven of the 34 treated counties are assessed twice. Under staggered timing such a regression uses counties already assessed as controls for counties assessed later, and its coefficient can mix effects of opposite sign (Goodman-Bacon, 2021). Estimating difference in differences one event at a time avoids that problem, but it weights every comparison county and every pre-period quarter equally and so requires the treated county's outcome to move parallel to the average comparison county's. Before their assessments, foreclosure filings in the treated counties were rising relative to the average annually assessing county (Figure 3). Equal weights would attribute that difference in trends to the assessment.

Synthetic difference in differences (Arkhangelsky et al., 2021) keeps the fixed-effects structure of difference in differences and chooses the weights. Comparison-county weights make the weighted comparison path parallel to the treated county's before the assessment, so the counterfactual is built from counties whose filings were moving as the treated county's were. Time weights favor the pre-period quarters that best predict the comparison counties' post-period, so a quarter deep in the foreclosure crisis, which says little about where filings were heading, contributes little to the baseline. County fixed effects absorb any constant difference in levels, which matters because the treated counties are larger than most of their comparison counties. For event e, with outcome Y_{it} the quarterly rate per 1,000 owner-occupied homes in county i, the estimate solves

$$\left(\hat{\tau}_e, \hat{\mu}, \hat{\alpha}, \hat{\beta}\right) \;=\; \underset{\tau,\mu,\alpha,\beta}{\arg\min} \sum_i \sum_t \left(Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau\right)^2 \hat{\omega}_i \, \hat{\lambda}_t, \tag{1}$$

where W_{it} equals one for the treated county after its assessment, α_i and β_t are county and quarter effects, and $\hat{\omega}_i$ and $\hat{\lambda}_t$ are the county and time weights, fitted separately for each event and outcome. The headline estimate averages the 45 events with equal weight,

$$\bar{\tau} \;=\; \tfrac{1}{45} \textstyle\sum_{e=1}^{45} \hat{\tau}_e. \tag{2}$$

The county weights solve

$$\left(\hat{\omega}_0, \hat{\omega}\right) = \underset{\omega_0 \in \mathbb{R},\, \omega \in \Omega}{\arg\min} \sum_{t \in \mathcal{T}^-} \Big(\omega_0 + \sum_j \omega_j Y_{jt} - Y_{0t}\Big)^2 + \zeta^2 T_{\text{pre}} \|\omega\|_2^2, \tag{3}$$

where Ω is the set of nonnegative weights that sum to one, j indexes comparison counties and 0 the treated county. The intercept ω_0 lets the weighted comparison path sit at a different level from the treated county's, so the weights reproduce its movement and leave its level to the county effect. The penalty ζ, set as in Arkhangelsky et al. (2021) to $T_{\text{post}}^{1/4}$ times the standard deviation of the comparison counties' quarter-to-quarter changes, spreads weight across many counties and makes the solution unique. In the main estimates the largest weight in the median fit is 5 to 7 percent.

The time weights solve the same problem across counties,

$$\left(\hat{\lambda}_0, \hat{\lambda}\right) = \underset{\lambda_0 \in \mathbb{R},\, \lambda \in \Lambda}{\arg\min} \sum_j \Big(\lambda_0 + \sum_{t \in \mathcal{T}^-} \lambda_t Y_{jt} - \tfrac{1}{T_{\text{post}}} \textstyle\sum_{t \in \mathcal{T}^+} Y_{jt}\Big)^2, \tag{4}$$

choosing the pre-period quarters that best predict the comparison counties' post-period average. They are fitted on the comparison counties alone, so the assessment shock cannot influence which quarters enter the baseline, and are then applied to the treated county as well. In the main estimates they concentrate on the quarters just before each assessment, with 49 to 69 percent on the last one on average.

Arkhangelsky et al. (2021) derive the estimator's properties under a model in which each county's outcome responds to shocks common to all counties, such as movements in mortgage rates, regional house prices and unemployment, with a sensitivity specific to the county,

$$Y_{it} \;=\; \gamma_i' \upsilon_t \;+\; \tau W_{it} \;+\; \varepsilon_{it}, \tag{5}$$

where υ_t collects the common shocks in quarter t and γ_i the county's exposure to each. County and quarter fixed effects are the special case in which every county responds one for one to a single common shock and differs only by a constant level. Under this model the estimator is consistent and asymptotically normal when the errors are independent across counties, when their serial correlation within a county is limited, and when the weighted comparison counties can reproduce the treated county's exposure to the common shocks. The last condition has a testable implication, since a weighted comparison with the same exposure tracks the treated county closely before the assessment. The setting adds two requirements. The comparison counties must be untreated, which annual assessment guarantees. The treated county must not be assessed again inside its own window, which the choice of windows guarantees.

Inference rests on the comparison counties. For each event the estimator is rerun with each comparison county in turn treated as if assessed, which gives a distribution of effects where no assessment occurred. Drawing one placebo effect per event and averaging gives a placebo average (Cavallo et al., 2013). The standard error of $\bar{\tau}$ is the standard deviation of 200,000 such averages, the placebo variance of Arkhangelsky et al. (2021) carried to an average across events, and significance uses normal critical values. Arkhangelsky et al. (2021) recommend the placebo variance when treated units are few, and each fit here has a single treated county. The placebo p is the share of the placebo averages at least as large in absolute value as $\bar{\tau}$. A second check moves the assessment to the midpoint of each pre-period, where none occurred, and estimates again.

The comparison counties are counties that meet the conditions of Section 5. A comparison county must be annually reassessed. The remaining screens require stable recording of sales and filings over the window, enough block groups to sort homes by value, and no quarter in which a value quartile is empty or thin. Each event draws on 90 to 107 counties that pass, and Table 2 charges every excluded county to the first condition it fails.

The events are countywide assessments in North Carolina between 2011 and 2018. Of the state's 124 assessments in those years, 104 are in counties where owner-occupied homes can be identified and sorted into value quartiles. An assessment enters only if its year was fixed before its window opened: every annual rate chart of the North Carolina Department of Revenue from the year the pre-period begins must name that year as the county's next assessment.[13] An assessment timed in response to conditions inside the window would confound the comparison, and the rule admits only assessments whose timing was settled before the window begins; 102 of the 104 pass. The treated county must then meet the same data conditions as the comparison counties, described below, which leaves 45 assessments in 34 counties (Table 2). Eleven of those counties assessed twice in that span. Twenty-one of the assessments closed four-year cycles, fifteen closed eight-year cycles, and the other nine closed cycles of five to seven years. They cluster in 2011 and 2012 and again from 2015 to 2017, with thirteen in 2017 alone.[14] Each window runs eight quarters before and eight after the assessment and contains no other assessment of the same county. Because cycle lengths differ across events, Section 6.2 also compares the estimates across the lengths of the cycles the assessments closed.

The county estimates describe the average owner. The same estimator, run separately on each value quartile's series with its own weights, shows how effects vary across the home value distribution. Quartiles are cut within each county on the block-group median of past sale prices, so each quartile compares homes

[13] The charts report each county's latest and next scheduled assessment year. Seventeen of the 45 assessments in the sample were rescheduled at some point in their cycles, and each had reached its final year by the chart of the year its pre-period began.

[14] In 2013 Mecklenburg reviewed its 2011 roll and revised about 18 percent of values, with refunds and back-taxes payable in 2015, inside the 2011 event's post-period, so the estimate for that event nets the original change against its partial reversal.

of similar value in the treated county with homes of similar value in the comparison counties. The quartiles warrant separate counterfactuals because cheap and expensive homes trade in different submarkets, with different buyers and financing (Landvoigt et al., 2015).

Each quartile's 45 estimates are averaged with equal weight, with the same placebo standard errors and placebo test as the county estimates. To test whether the effect differs across quartiles, I compare each of the three cheaper quartiles, and their average, with the most expensive. Each comparison is the average over the 45 assessments of the difference between that assessment's quartile estimates,

$$\bar{\delta}_q = \frac{1}{45}\sum_e \left(\hat{\tau}_{eq} - \hat{\tau}_{e4}\right), \quad q = 1, 2, 3, \qquad \bar{\delta}_{1\text{–}3} = \frac{1}{45}\sum_e \left(\frac{\hat{\tau}_{e1} + \hat{\tau}_{e2} + \hat{\tau}_{e3}}{3} - \hat{\tau}_{e4}\right), \tag{6}$$

where $\hat{\tau}_{eq}$ is the synthetic difference-in-differences estimate for assessment e in quartile q and the fourth quartile is the most expensive, so the comparison requires no further regression and gives each assessment equal weight, as the main averages do. Its placebo test draws, for each assessment, one comparison county and treats it as the assessed county in every quartile at once, so the placebo contrasts keep the correlation between the quartile estimates.

There are three outcomes. A foreclosure filing is the recorded start of a foreclosure proceeding against the home. A completed foreclosure is the transfer of title to the lender or a buyer at the foreclosure sale. An underwater sale is a sale for less than the outstanding mortgage. Each is a quarterly rate per 1,000 owner-occupied homes, seasonally adjusted within county.

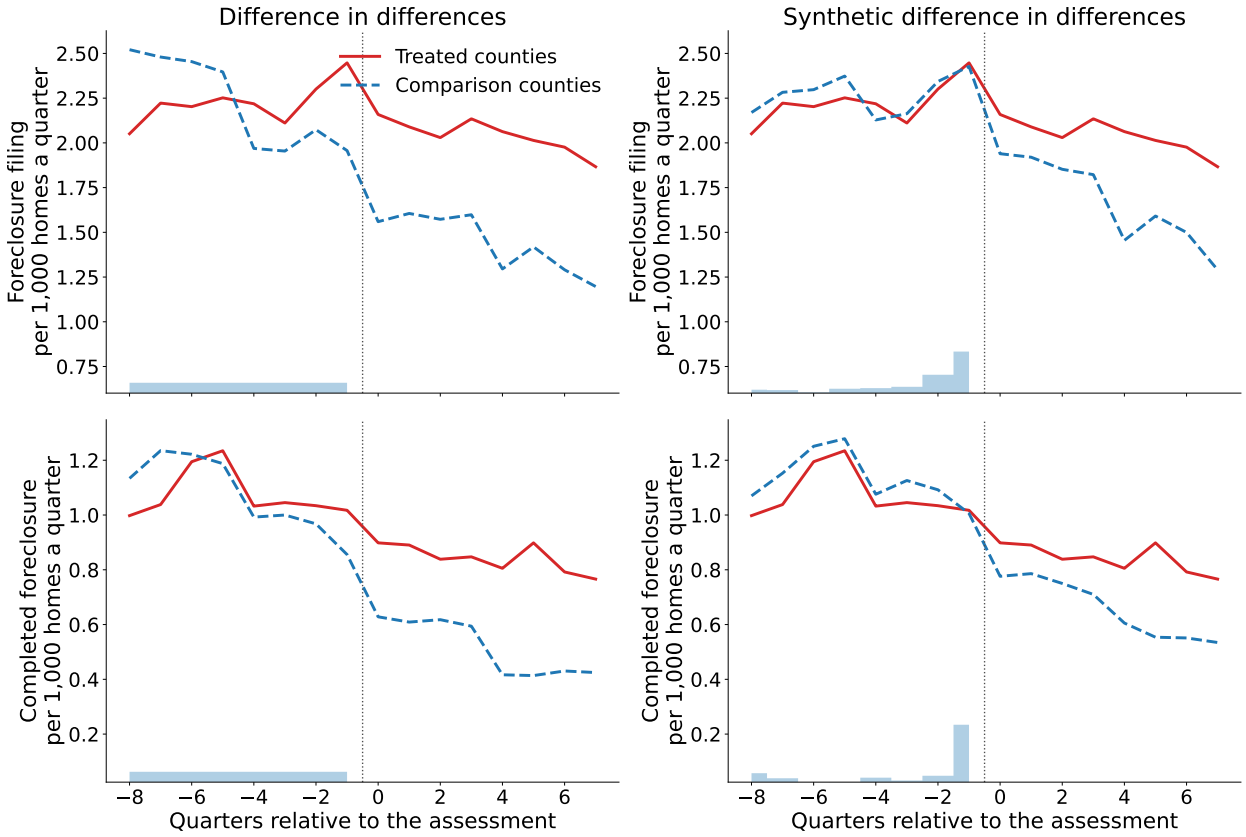


Figure 3: Treated counties and their comparisons by quarter, difference in differences and synthetic difference in differences.

Notes. Each line averages the 45 assessments in quarters relative to the assessment. The left column compares the treated counties with the equal-weighted mean of their comparison counties, and the right column with their weighted synthetic comparisons; each comparison is shifted by its time-weighted pre-period gap, the level difference the county effect absorbs. Bars are the time weights, equal under difference in differences and averaged over events under synthetic difference in differences. Outcomes are quarterly rates per 1,000 owner-occupied homes. The dotted line marks the assessment.

Figure 3 shows both comparisons quarter by quarter. Against the average comparison county, the treated counties' filing rate rose before the assessment: the gap between them is 0.26 to 0.96 per 1,000 homes a quarter lower in the two years before the assessment than in the quarter just before it, so difference in differences counts that rise as part of the effect. Synthetic difference in differences keeps every pre-period filing gap within 0.15 per 1,000 of the last pre-period quarter. Its post-period gap is 0.20 in the quarter of the assessment and between 0.29 and 0.59 from the fourth quarter on. Completed foreclosures show a smaller pre-period rise against the average county, 0.10 to 0.36 per 1,000, and underwater sales almost none; under synthetic difference in differences both stay within 0.13 per 1,000 of the last pre-period quarter and separate after the assessment, most from the fifth quarter on, as filings reach completion.

5 Data

The deed and assessment records come from the Zillow Transaction and Assessment Database (ZTRAX, ICPSR Study 39652). Its transaction files record every deed, mortgage and foreclosure notice filed with the county recorder since about 1990, with the document type, recording date, sale price and loan amount. Its assessment files are a recent cross section of every parcel, with physical characteristics, location and the assessor's record of owner-occupancy. I link the two through the parcel identifier and express every dollar amount in 2025 dollars using the annual-average consumer price index for all urban consumers.[15]

The sample is owner-occupied homes, since rentals and second homes are not held by the households whose mortgage payments an assessment changes. In most counties owner-occupancy comes from the assessor's occupancy field in ZTRAX. In Florida it comes from the state's statutory property rolls, which record the homestead exemption that only a resident owner can claim.[16] Counties with no source from which owner-occupants can be identified are left out of the comparison pool.

A foreclosure filing is a recorded notice that opens a foreclosure, a notice of default, notice of sale or lis pendens, identified by the same rule in every state. A completed foreclosure is a deed that transfers the home at a foreclosure sale, a foreclosure, trustee's or sheriff's deed or a certificate of sale, at any recorded price. An underwater sale is an arm's-length sale below the home's estimated payoff, the outstanding mortgage debt plus arrears and about 6 percent in selling costs. The debt amortizes each recorded loan over 30 years at 5 percent, which overstates it for owners who prepaid. Guilford's recorded filings are incomplete before 2012, at 4 and 28 percent of the state court count in 2010 and 2011, so in every county a completed foreclosure with no filing in the three years before it is assigned one at the county's median gap between filing and completion. Each outcome is the number of owner-occupied homes with the event in a quarter per 1,000 owner-occupied homes, seasonally adjusted within county.

A filing is the first recorded step of every mortgage foreclosure, and Figure A3 shows how filings end. It follows each filing on an owner-occupied home in the five North Carolina counties between 2011 and 2018 for eight quarters and records the first document that follows it. About 60 percent of filings reach a recorded outcome within two years: 36 percent end in a completed foreclosure, 11 percent in a sale with equity, 9 percent in a refinance, 3 percent in an underwater sale, and under 1 percent in a deed in lieu or a tax deed. The remaining 40 percent leave nothing on record, which covers cures, loan modifications and cases still pending. Owners without a mortgage can lose a home to a tax deed with no filing (Figure A2), a path the outcomes here do not follow.

[15] `https://fred.stlouisfed.org/series/CPIAUCSL`

[16] Michigan's ZTRAX occupancy field does not distinguish owners from absentee owners, so there a home is owner-occupied when the owner's mailing address in the county's parcel records matches the property address.

To describe how effects vary across the home value distribution, I sort owner-occupied homes into value quartiles within each county. Each home that sold at arm's length between 1996 and the anchor year is valued at the median of its sale prices in 2025 dollars, and each census block group with at least 20 such homes is given the median of their values. Every owner-occupied home in the block group carries that value, sold or not, and quartiles are cut within county. The anchor year closes before each event's pre-period opens, so the sort uses no price from the window being studied. Sorting on the block group places homes that have not sold recently, which a sort on each home's own sale would leave out, and in Mecklenburg it places 94.5 percent of owner-occupied homes in a quartile. It also averages away the transitory part of any single sale, which would otherwise revert toward the middle of the distribution and manufacture part of the gradient being measured. The quartiles therefore describe the cheapest quarter of neighborhoods within a county.

Tax liabilities come from county and state property rolls and are used only in the tax liability exercise of Section 6.3. Mecklenburg publishes a parcel-by-year record of assessed values covering 2001 and 2004 through 2023,[17][18] and Guilford publishes annual rolls from 2008 to 2025. Wake and Durham rolls give the same record, and for the other ten counties assessed values come from county roll files and dated statewide parcel snapshots. A North Carolina tax liability is the parcel's assessed value times its combined county, municipal and district rate, with rates from the North Carolina Department of Revenue and county budget ordinances. Where a county also publishes the tax liabilities it mailed, liabilities computed this way match them within 1 percent for 98 to 99.8 percent of parcels. For the comparison counties, Florida's Department of Revenue publishes statutory rolls of every parcel from 2002 to 2023. A Florida tax liability is the parcel's school taxable value times the county's total millage. I match these rolls to the ZTRAX parcels in 64 of Florida's 67 counties.

The 45 assessments, their dates and the length of the cycle each closed come from the North Carolina Department of Revenue's record of county assessment years. Table 1 describes them by cycle length. The treated counties hold between 11,000 and 281,000 owner-occupied homes, 34,000 in the median event and 2.25 million across all 45. Before the assessments, filings averaged 2.23 per 1,000 homes a quarter, completed foreclosures 1.07 and underwater sales 0.59, below the comparison pool's 2.95, 1.57 and 1.13. The county effects absorb that difference in level. Each event draws on 90 to 107 eligible comparison counties.

[17] The county archive holds no 2003 roll, and its 2002 file was written in April 2003 already carrying the 2003 reassessment, so neither year yields a tax liability.

[18] `https://mecksusftp001.blob.core.windows.net/gis/OpenMapping/index.html`

Table 1: The 45 assessments, by length of cycle.

				Treated counties		Comparison counties		
Cycle	Events	Counties	Homes	Filing	Fore.	Filing	Fore.	Counties
4 years	21	14	39,161	2.41	1.20	3.05	1.65	90–107
5 years	5	4	58,299	2.17	1.03	3.36	1.72	90–100
6–7 years	4	4	17,906	2.43	0.78	2.81	1.32	97–107
8 years	15	15	30,801	1.92	0.99	2.71	1.49	90–100
All	45	34	33,680	2.23	1.07	2.95	1.57	90–107

Notes. Cycle is the years since the county's previous assessment. Events counts assessments and Counties the distinct treated counties in each row; a county that assessed twice can appear in two rows. Homes is the median number of owner-occupied homes per event. Rates are quarterly rates per 1,000 owner-occupied homes over the eight quarters before the assessment, averaged over events. Comparison rates average each event's eligible comparison counties with equal weight; the last column is the range of eligible comparison counties per event.

Selection into the comparison unit. Every county with identifiable owner-occupants and a usable record of sales enters the candidate set, and four conditions then decide, event by event, which can serve as a comparison. First, the county must assess annually, so it absorbs no discrete correction of its own inside the event's window. Second, its deed recordings must track its housing market over the window, so its pre-period sales must reach at least 35 percent of its own best four-year stretch.[19] Third, the county must have at least 20 block groups with enough sales to receive a value, together holding at least 60 percent of its owner-occupied stock, so that its homes can be sorted by value. Fourth, every value quartile must record at least 10 sales in every quarter of the window, so that no quartile-quarter rate rests on a handful of sales. The treated events must meet the second, third and fourth conditions as well, so no treated estimate rests on thinner data than its comparison; 45 of the 102 assessments whose timing was fixed in advance pass.

Nine states assess annually, containing 745 counties, and I extract all of them (Table 2). A median of 276 across the 45 events have identifiable owner-occupants and can be sorted into value quartiles. Stable coverage across the window leaves a median of 222, enough block groups to sort homes by value 145, and no thin or empty quartile-quarter 97. Between 90 and 107 counties remain, depending on the event. The same conditions applied to the treated assessments leave 90, 68 and 45 of the 102 assessments whose timing was fixed in advance.

The screen governs the placebo distribution as well as the comparison pool. Inference compares the treated county's divergence after its assessment with the divergence of each comparison county treated as if it had been assessed. A county whose deed record is filling in shows a compressed path before the assessment and a rising one after it, so it registers a large apparent effect for reasons unrelated to any assessment and displaces the treated county in the ranking. Excluding such a county from the weights while retaining it as

[19] Coverage in several states improves across the sample for administrative reasons, which produces a rising sales series that has nothing to do with housing. Measuring against the county's own best stretch keeps counties whose sales fell during the crisis.

a placebo would leave the comparison distorted.

Each event admits between 90 and 107 comparison counties (Table 2), but the estimator does not use them equally. Weights are fitted separately for every event and outcome, so each estimate has its own comparison group, and Appendix Figure A1 and Table A1 report which counties those are. The filing fits draw on 22 to 95 counties per event, and the largest single county carries between 2.4 and 14.1 percent of an event's fit. Across the 45 events, all 127 eligible counties carry filing weight in at least one fit.

Table 2: Screening of comparison counties and treated assessments.

A. Comparison counties	Median across events	Fewest	Most
Annually assessing counties, nine states	745	745	745
Owner-occupancy and value quartiles available	276	253	285
Coverage stable across the window	222	215	238
Enough distinct areas to sort on	145	143	158
No thin or empty quartile-quarters	**97**	90	107
B. Treated assessments	Assessments		
North Carolina assessments, 2011–2018	124		
Owner-occupancy and value quartiles available	104		
Year announced before the window opens	102		
Coverage stable across the window	90		
Enough distinct areas to sort on	68		
No thin or empty quartile-quarters	**45**		

Notes. Panel A counts the comparison counties remaining after each condition, as the median across the 45 events, with the fewest and most. Panel B counts the North Carolina assessments of 2011–2018 remaining after each condition. A county or assessment failing more than one condition is charged to the first.

6 Results

6.1 Synthetic difference-in-differences estimates

Figure 4 sets the treated counties beside their synthetic comparisons, averaged over the 45 assessments. The weights are chosen to make the two paths parallel before the assessment, and the time weights beneath each panel fall mostly on the last pre-period quarter, which carries half to two thirds of the weight. After the assessment, filings in the treated counties stay between 1.9 and 2.2 per 1,000 homes a quarter while the synthetic comparison falls from 1.9 to 1.3. Completed foreclosures and underwater sales separate the same way, and all three gaps widen after the first year.

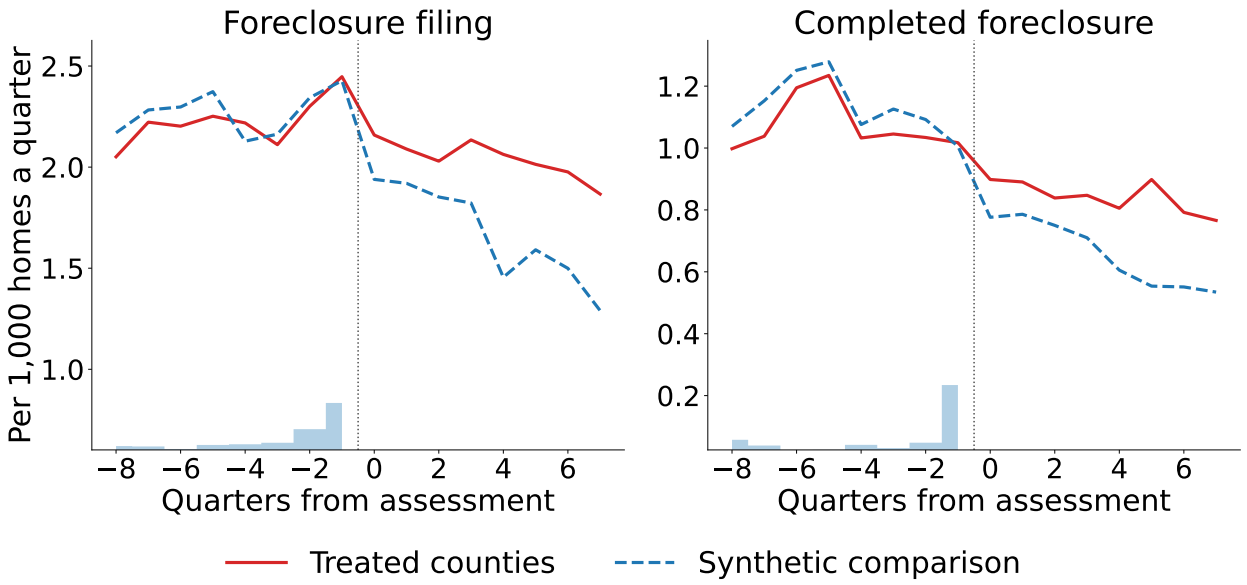


Figure 4: Treated counties and their synthetic comparisons, average over 45 assessments.

Notes. Each line averages the 45 assessments in quarters relative to the assessment. The synthetic comparison applies each event's comparison-county weights for that outcome and is shifted by the time-weighted pre-period gap, the level difference the county effect absorbs. Bars are the time weights, averaged over events. Outcomes are quarterly rates per 1,000 owner-occupied homes. The dotted line marks the assessment.

Table 3 reports the average effect of the 45 assessments on all owner-occupied homes.

Table 3: The effect of an assessment, average over 45 events, all homes.

	Foreclosure filing	Completed foreclosure
	0.37***	0.18**
	(0.13)	(0.07)
Pre-period base	2.23	1.07
Relative to base	17%	17%
Total over 8 quarters	2.96	1.47
Events (counties)	45 (34)	
Treated homes	2,250,509	
Comparison counties per event	90–107	

Notes. Each assessment and outcome is estimated separately by synthetic difference in differences, with its own comparison-county and time weights. Each column averages the 45 assessments' estimates with equal weight. Outcomes are quarterly rates per 1,000 owner-occupied homes, eight quarters before and after. Relative to base divides the effect by the pre-period rate, and the total over 8 quarters multiplies it by eight. Standard errors are the standard deviation of averages from comparison counties treated as if assessed, with normal critical values; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

An assessment raises the probability of every stage of mortgage distress, as the first prediction of Section 3 implies (Table 3). In the two years after an assessment, the quarterly foreclosure filing rate rises by 0.37 per 1,000 owner-occupied homes, 17 percent of its pre-period rate of 2.23. Over the eight quarters that follow, this accumulates to about 3.0 additional filings per 1,000 owner-occupied homes, or about 100 filings

in the median treated county of 34,000 homes. Completed foreclosures rise by 0.18 per 1,000 a quarter and underwater sales by 0.09, 17 and 16 percent of their bases. These amount to 1.5 and 0.8 homes per 1,000 over two years, about 50 foreclosures and 25 underwater sales in the median county. Across all 45 assessments, which together cover 2.25 million owner-occupied homes, these rates imply about 6,700 additional foreclosure filings, 3,300 additional completed foreclosures and 1,700 additional underwater sales over the two years after the assessments. The additional completed foreclosures equal about half the additional filings, so the filings an assessment adds end in foreclosure at least as often as ordinary ones, about a third of which complete within two years (Figure A3).

All three estimates are distinguishable from zero against averages built from comparison counties treated as if assessed, with p-values of 0.005 for filings, 0.013 for completed foreclosures and 0.008 for underwater sales.

The synthetic comparisons track the treated counties closely before each assessment. The root mean squared gap over the pre-period averages 10 percent of the base for filings, 11 percent for completed foreclosures and 14 percent for underwater sales. The county weights are dispersed: the largest weight in the median fit is 5 to 7 percent, equivalent to about 40 equally weighted comparison counties, so no single county drives an estimate. The time weights concentrate on the quarters just before each assessment, with 49 to 69 percent on average on the final pre-period quarter, so each estimate compares the post-period with the treated county's most recent trajectory. Difference in differences, which weights every comparison county and pre-period quarter equally, gives larger estimates for all three outcomes, 0.60 per 1,000 for filings, 0.33 for completed foreclosures and 0.16 for underwater sales. The larger estimates are consistent with distress in the treated counties already rising relative to the average comparison county, the trend the synthetic weights remove.

6.2 Does the effect grow with the length of the cycle?

Longer cycles leave more appreciation untaxed, so the assessment that ends one should correct a larger gap. Figure 5 groups the 45 assessments by the years since the county's previous assessment. Filing effects average 0.22 per 1,000 homes a quarter after the 21 four-year cycles and 0.49 after the 15 eight-year cycles. Across assessments, the fitted slope is 0.055 per 1,000 a quarter for each additional year (standard error 0.072), about 2 percent of the pre-period rate, or 0.4 filings per 1,000 homes over the eight post-period quarters. The slope for completed foreclosures is 0.034 (standard error 0.040), 3 percent of its base. The slope for underwater sales is 0.047 (standard error 0.019), 8 percent of its base, and is the only one distinguishable from zero; each additional year of cycle adds 0.4 underwater sales per 1,000 homes over two years. All three

slopes stay positive among the 28 assessments of 2014 to 2018, which leaves out the foreclosure crisis, and there underwater sales rise by 0.24 per 1,000 a quarter after eight-year cycles against −0.03 after four-year cycles. Cycle length varies across counties more than within them, so the comparison cannot separate it from differences across counties and housing markets, and should be read as descriptive.

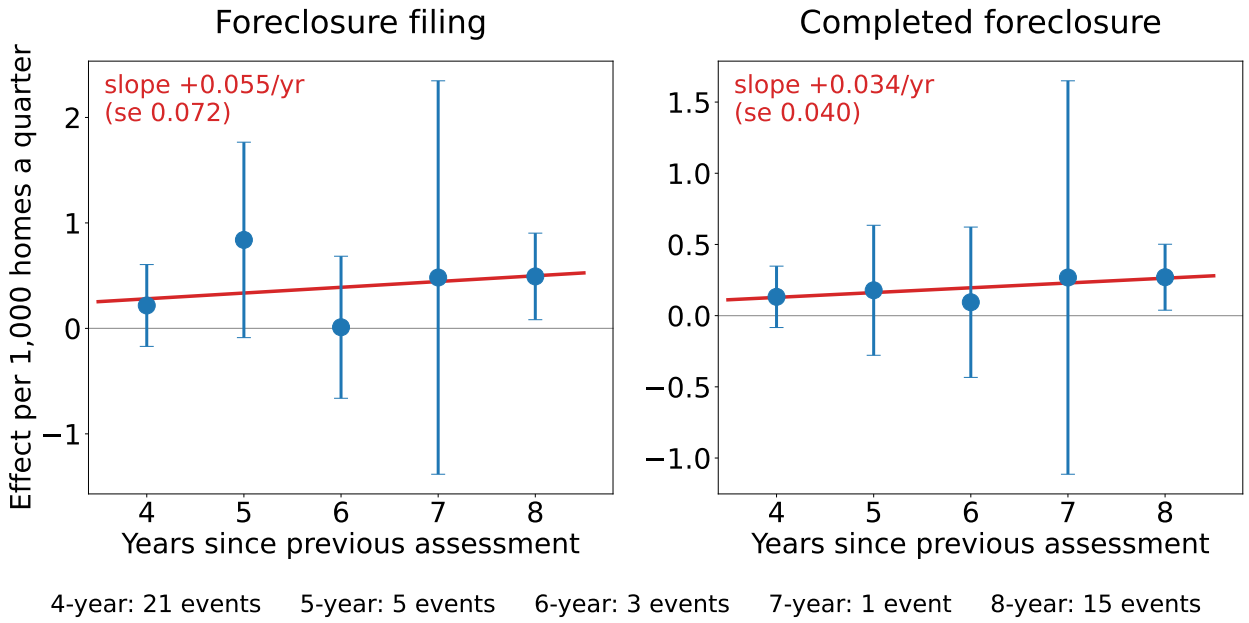


Figure 5: The effect of an assessment by the length of the cycle it closed.

Notes. Each point averages, with equal weight, the synthetic difference-in-differences effects on all owner-occupied homes of the assessments that closed a cycle of that length, over eight post-period quarters, per 1,000 homes a quarter. Bars are 95 percent intervals from the placebo standard error of that average. The red line is the least-squares fit across the 45 assessments; its standard error is the spread of slopes fitted to placebo draws, one comparison county per assessment. The number of assessments at each cycle length is listed below the panels.

6.3 Scaling the Effect by the tax liability

This section asks whether the effect of an assessment scales with the tax change it produced. The test requires the tax liability on both sides of the comparison. In 18 of the 45 assessments, in 14 counties, county rolls record each parcel's assessed value and tax rate in every year of the window, and Florida's statutory rolls record the same for its counties, so these 18 assessments are estimated again against Florida comparison counties only. The approach follows studies that identify the effect of payment size on default from payment changes set by rule (Fuster and Willen, 2017; Anderson and Dokko, 2016): the assessment date is fixed by statute, and the tax change it produces differs across assessments.

For each event e and value quartile q, the synthetic difference-in-differences estimate $\hat{\tau}_{eq}$ is the change in the quarterly rate after the assessment in the North Carolina county relative to its weighted Florida donors.

The tax change is the same comparison applied to the tax liability,

$$\Delta b_{eq} = \left(\bar{b}_{eq}^{\text{post}} - \bar{b}_{eq}^{\text{pre}}\right)_{\text{NC}} - \sum_{j} \omega_{jeq} \left(\bar{b}_{jq}^{\text{post}} - \bar{b}_{jq}^{\text{pre}}\right), \tag{7}$$

where $\bar{b}$ is the mean log tax over the eight quarters before or after the assessment and ω_{jeq} is the weight on Florida donor j. Dividing the effect by the tax change gives the change in the quarterly rate per unit of tax change, with the assessment acting as an instrument for the tax. The 18 assessments are pooled as the ratio of averages,

$$\hat{\beta}_q = \frac{\frac{1}{18}\sum_e \hat{\tau}_{eq}}{\frac{1}{18}\sum_e \Delta b_{eq}}, \tag{8}$$

and I report $\hat{\beta}_q \times 0.01$, the change per 1 percent increase in the tax liability. North Carolina tax liabilities are assessed value times the parcel's combined county, municipal and district rate; Florida tax liabilities are school taxable value times the county's total millage. The tax change is measured with each outcome's own donor weights, so it differs slightly across outcomes. Standard errors are the standard deviation of placebo averages, each built from one randomly drawn Florida comparison county's placebo estimate per event, divided by the average tax change. The estimate attributes the whole effect of the assessment to the tax liability, so any effect that runs through the value the county states for the home is counted as a tax liability effect.

Table 4: Assessment effects scaled by the tax-tax change, 18 North Carolina assessments against Florida comparison counties.

	Foreclosure filing	Completed foreclosure
Per 1% bill increase	−0.13	0.11**
	(0.09)	(0.05)
Bill change vs. Florida	0.8%	2.1%
Pre-period base	2.29	1.16
Per 1%, relative to base	−5.6%	9.2%
Per 1%, total over 8 quarters	−1.02	0.86
Events (counties)	18 (14)	
Florida comparison counties per event	36–42	

Notes. Each entry is the average synthetic difference-in-differences estimate over the 18 assessments, divided by the average tax change (equation 8), per 1,000 owner-occupied homes a quarter per 1 percent tax liability increase. The tax change is the post-minus-pre change in the mean log tax liability, North Carolina less the weighted Florida comparison counties, times 100. Standard errors are the spread of placebo averages, with the tax change treated as known. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 reports the estimates. Completed foreclosures respond to the size of the tax liability. Each additional percent of tax liability increase relative to Florida raises the quarterly completed foreclosure rate by 0.11 per 1,000 owner-occupied homes (standard error 0.05), 9 percent of its pre-period rate of 1.16, which accumulates to 0.9 completed foreclosures per 1,000 homes over two years. Underwater sales rise by 0.04

per 1,000 per percent (standard error 0.04), 5 percent of their base, an estimate indistinguishable from zero. Foreclosure filings show no such response; their estimate is −0.13 per 1,000 per percent (standard error 0.09). The tax liability increase against Florida averages about 2 percent across these assessments, so an effect per percent stays within the range of tax changes the assessments delivered. The estimate is the average effect of an assessment expressed per percent of the average tax change, and it does not imply that assessments with larger tax liability increases produced larger effects: the 2011–2013 assessments had the largest increases relative to Florida, because Florida tax liabilities fell in those years, and among the smallest effects.

7 Estimates of the avoided distress under annual assessment

Long assessment cycles impose a cost that can be measured directly. Because every comparison county assesses its homes annually, the estimated effect of an assessment is the additional distress it causes relative to annual assessment, and multiplying that effect by the homes assessed gives the distress an annual schedule would have avoided. Across the 45 assessments, which cover 2.25 million owner-occupied homes, the quarterly filing rate rises by 0.50 per 1,000 homes and the completed foreclosure rate by 0.17 per 1,000, or 21 and 14 percent of their pre-period rates. Over the two years after these assessments, annual assessment would therefore have avoided about 9,000 foreclosure filings and 3,100 completed foreclosures. With a home-weighted average cycle of 6.1 years, that amounts to about 0.65 fewer filings and 0.22 fewer completed foreclosures per 1,000 homes each year, or about 1,100 filings and 375 completed foreclosures a year across the 34 counties.

7.1 Heterogeneity by Home Value Quartile

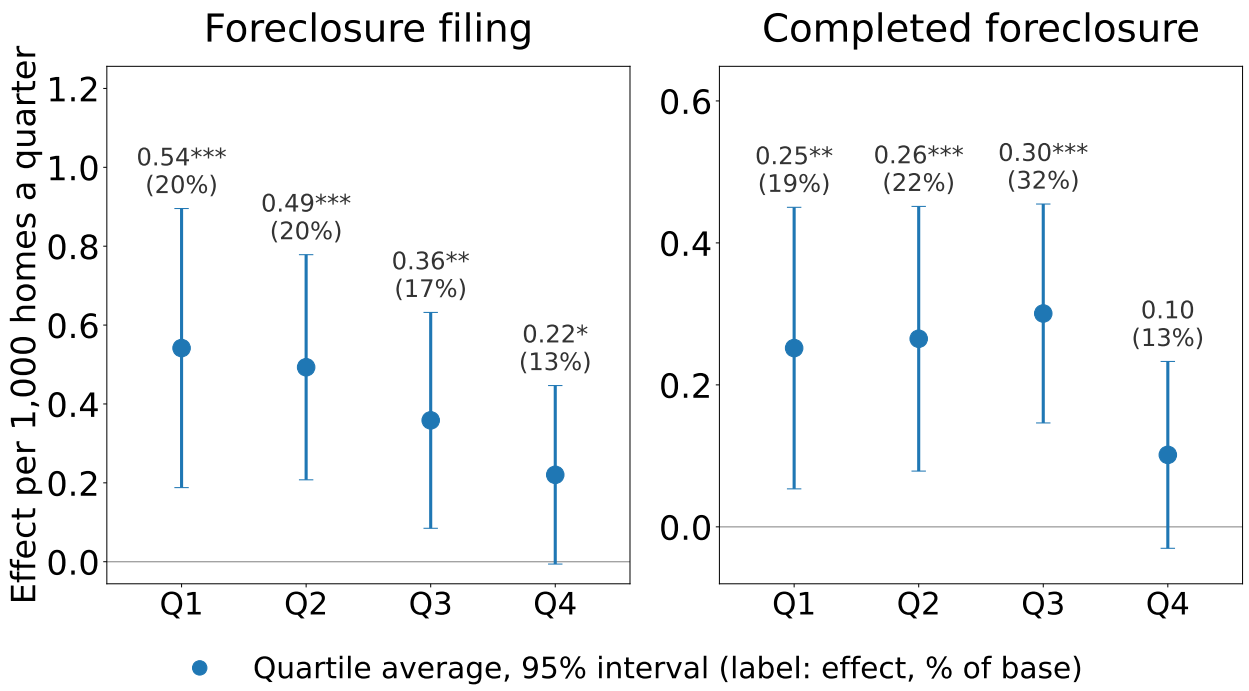


Figure 6: The effect of an assessment by home-value quartile, average over 45 events.

Notes. Each point averages, with equal weight, the 45 synthetic difference-in-differences estimates for that quartile, each with its own comparison-county and time weights. Bars are 95 percent intervals from the placebo standard error, the standard deviation of averages from comparison counties treated as if assessed. Labels give the effect and, in parentheses, the effect relative to the quartile's pre-period rate; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Outcomes are quarterly rates per 1,000 owner-occupied homes in the quartile, eight quarters before and after. Quartiles are cut within county on the block-group median of past sale prices; Q1 is the least expensive.

The bottom three value quartiles drive the average effect (Figure 6). Foreclosure filings decline steadily with home value. The quarterly filing rate rises by 0.54 per 1,000 homes in the cheapest quartile and by 0.49, 0.36 and 0.22 in the higher ones, which accumulate to 4.3, 3.9, 2.9 and 1.8 additional filings per 1,000 homes over two years. The decline holds relative to each quartile's own pre-period rate, from 20 percent in the two cheapest quartiles to 17 percent in the third and 13 percent in the most expensive, so it does not simply reflect the higher baseline filing rate of cheaper homes. Completed foreclosures and underwater sales rise by similar amounts across the bottom three quartiles, 0.25 to 0.30 per 1,000 a quarter for completed foreclosures and 0.11 to 0.17 for underwater sales, and by about 0.10 for each in the most expensive. The most expensive quartile therefore shows the smallest effect on every outcome, both in levels and relative to its base, and its completed foreclosure estimate is the only quartile estimate indistinguishable from zero. Figure 7 tests these differences against the most expensive quartile. For filings, the average of the three cheaper quartiles exceeds the most expensive by 0.24 per 1,000 homes a quarter ($p < 0.01$), and the cheapest and second quartiles each exceed it by 0.32 and 0.27 ($p = 0.02$ and $p < 0.01$), while the third quartile's margin of 0.14 is not

distinguishable from zero. For completed foreclosures the three cheaper quartiles exceed the most expensive by 0.17 on average ($p < 0.01$), by 0.15 to 0.20 each, significant for the second and third quartiles and at the 10 percent level for the cheapest. Underwater sales do not differ across quartiles. The decline in filings from one quartile to the next is not itself distinguishable from zero; what the data support is that the top quartile responds least. Larger effects below the top quartile are consistent with more of those owners being close to default, with thinner equity and less liquid savings to absorb a higher payment.

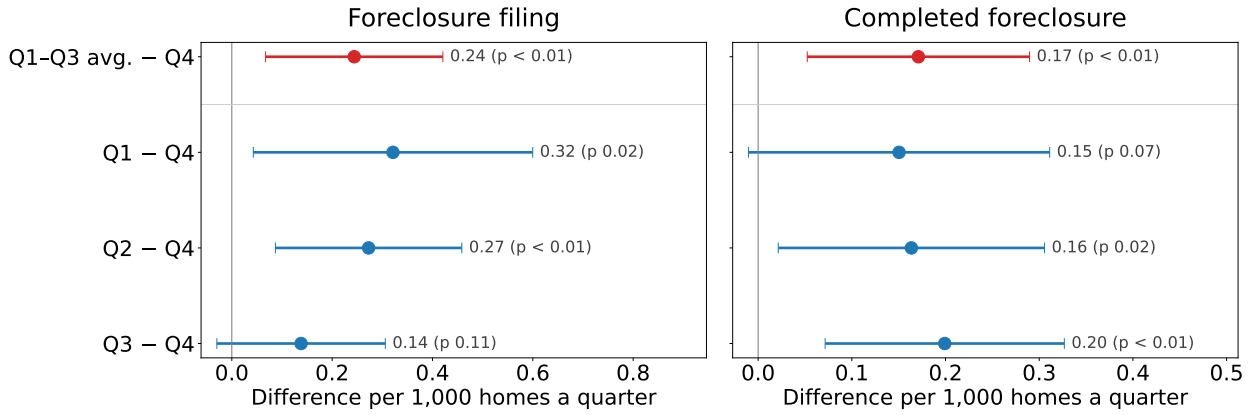


Figure 7: Differences in the effect of an assessment between value quartiles, relative to the most expensive quartile.

Notes. Each point is the equal-weight average over the 45 assessments of the difference between quartile estimates, per 1,000 owner-occupied homes a quarter, with its 95 percent placebo interval and p-value. The top row compares the average of the three cheaper quartiles with the most expensive. Each placebo draw treats one comparison county per assessment as the assessed county in every quartile, so the intervals account for the correlation between quartile estimates.

8 Conclusion

This study finds that multi-year reassessment cycles raise foreclosure filing and completion rates in the two years after the correction. I find that filings and completions rise by 17 percent of their pre-period windows, translating to 2.5 and 2.9 additional filings and completions per 1,000 owner-occupied homes in the studied counties. Pooling estimates by cycle length shows descriptively that an additonal year between reassessments is associated with a two and three percent increase in filings and completions, respectively, and that the longest cycles (eight years) drive the average reassessment effect. Wald ratios that divide the average tax change by the average effect for each event suggest that a one percent increase in the tax bill at a reassessment suggest that each additional percent increase in the tax liability raises filings and completions by three and nine percent, respectively. Estimating the effects of a reassessment by value quartile suggest that the owners of the most expensive quartile of homes largely absorb the tax change while the owners of lower value homes experience more filings and completions.

The central question in the paper asks whether the tax savings accrued by owners of fast-appreciating

homes are sufficient to offset the costs of steep changes in their tax liabilities after a reassessment. Postassessment increases in foreclosure filings and completions suggest that the tax savings are not sufficient to offset the liquidity strain owners face after reassessment. A back-of-the-envelope calculation suggests that under annual assessment, the studied counties would have avoided 3,100 foreclosures across the 2.5 million owner occupied homes in the sample. The costs faced by these homeowners should be weighed against the administrative costs of conducting annual assessments to answer whether these counties should adopt annual cycles.

The counties studied here reflect assessment statutes common to many counties across the U.S.: more than half of U.S. homes exist in counties with multi-year cycles. However, North Carolina homeowners do not receive the same protections held by many homeowners in other states, namely homestead exemptions. Therefore, whether the effects of this study can be generalized to counties outside of North Carolina is an open question.

A Appendix

A.1 Comparison-county weights

Each assessment and outcome has its own comparison-county weights. Figure A1 shows each comparison county's foreclosure filing weight averaged over the 45 assessments, and Table A1 lists each county's average weight for each outcome. All 127 comparison counties carry weight in at least one fit, no county averages more than 2.1 percent of the filing comparison, and the ten largest together hold 18 percent. Florida supplies about a third of the weight, Michigan a fifth, and Alabama, Minnesota and Georgia most of the rest.

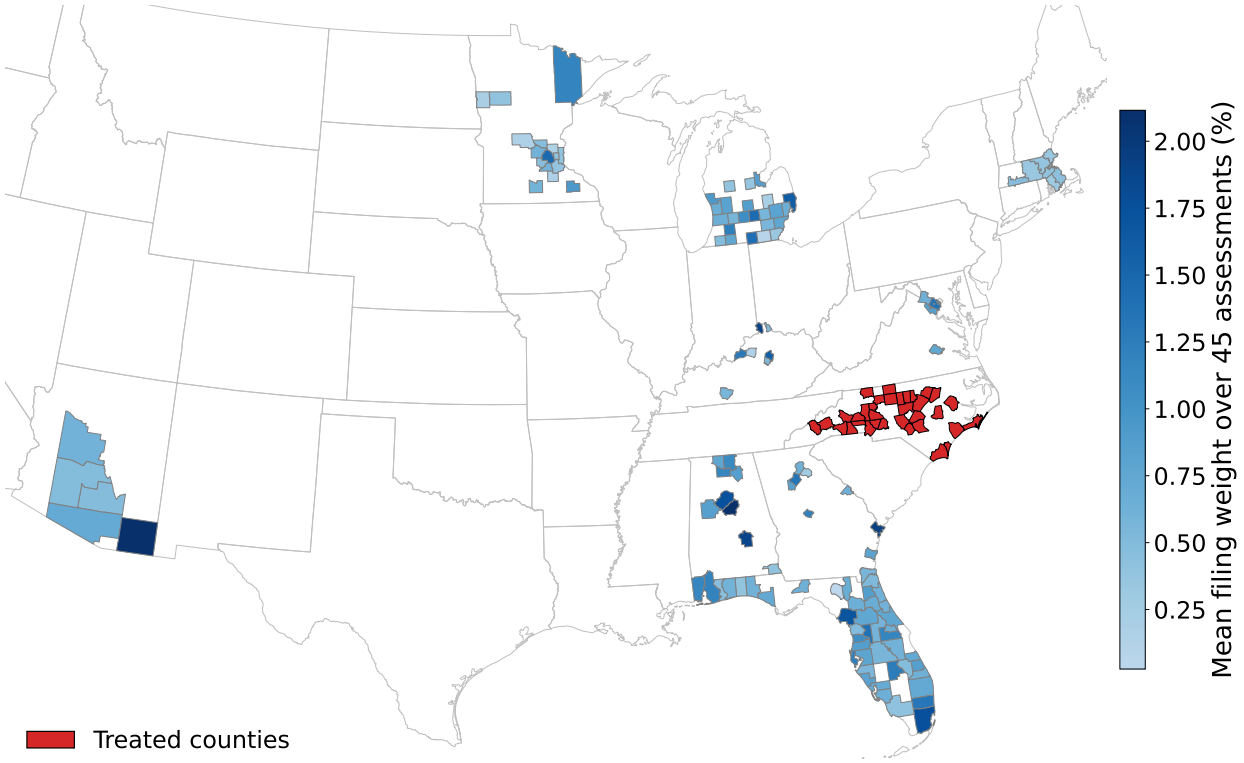


Figure A1: Comparison-county weights for foreclosure filings, average over 45 assessments.

Notes. Each comparison county is shaded by its foreclosure filing weight averaged over the 45 assessments, in percent; each assessment's weights sum to one, so a county's shading is its share of the whole filing comparison, and a county outside an assessment's pool counts as zero there. Treated counties are in red.

Table A1: Comparison-county weights by outcome, average over 45 assessments.

Notes. Each county's weight averaged over the 45 assessments, in percent, for foreclosure filings and completed foreclosures; a county outside an assessment's pool counts as zero there. The last column counts the assessments in which the county carries filing weight. Counties are sorted by mean filing weight.

	Mean weight over 45 assessments (%)		Assessments
County	Filing	Completed foreclosure	with filing weight
Shelby, AL	2.12	0.65	32
Cochise, AZ	2.11	1.57	30
Chatham, GA	1.94	2.13	35
Boone, KY	1.91	1.22	33
Montgomery, AL	1.88	0.77	40
Miami-Dade, FL	1.76	1.01	28
Jefferson, AL	1.75	1.36	33
Levy, FL	1.70	0.89	28
Fayette, KY	1.60	0.00	39
St. Clair, MI	1.60	0.41	15
Hennepin, MN	1.48	1.98	39
Ingham, MI	1.43	1.32	30
Hillsdale, MI	1.41	0.77	19
Sumter, FL	1.38	1.11	38
Fairfax, VA	1.36	1.13	31
Gwinnett, GA	1.36	1.16	28
Broward, FL	1.33	0.65	26
Jefferson, KY	1.30	1.24	30
Highlands, FL	1.27	0.94	30
Kalamazoo, MI	1.24	0.97	32
Pinellas, FL	1.23	0.88	32
Bibb, GA	1.21	3.14	28
Baldwin, AL	1.19	0.84	28
St. Louis, MN	1.18	1.09	38
Orange, FL	1.17	0.67	33
Mobile, AL	1.16	0.58	24
Morgan, AL	1.16	0.83	25
Alexandria, VA	1.14	1.23	21
Madison, AL	1.11	0.58	25
Eaton, MI	1.07	1.14	30
Hernando, FL	1.06	0.92	32
Arlington, VA	0.99	1.26	32
Muskegon, MI	0.94	0.79	31
Olmsted, MN	0.93	1.24	27
Bay, MI	0.88	0.47	28
Clay, FL	0.88	0.52	22
Marshall, AL	0.87	0.49	18
Clinton, MI	0.87	0.77	27
Pasco, FL	0.86	0.89	31
St. Lucie, FL	0.86	1.09	24

Table A1: Comparison-county weights by outcome, average over 45 assessments.

Notes. Each county's weight averaged over the 45 assessments, in percent, for foreclosure filings and completed foreclosures; a county outside an assessment's pool counts as zero there. The last column counts the assessments in which the county carries filing weight. Counties are sorted by mean filing weight.

	Mean weight over 45 assessments (%)		Assessments
County	Filing	Completed foreclosure	with filing weight
DeKalb, GA	0.86	0.37	8
Prince William, VA	0.86	1.30	29
Tuscaloosa, AL	0.85	0.60	17
Sarasota, FL	0.82	0.82	32
Kent, MI	0.81	1.65	29
Oakland, MI	0.81	0.71	30
Glynn, GA	0.80	1.57	22
Hillsborough, FL	0.79	0.93	27
Limestone, AL	0.78	0.50	12
Wayne, MI	0.77	0.75	28
Volusia, FL	0.77	0.99	25
St. Joseph, MI	0.77	1.29	19
Marion, FL	0.76	0.48	28
Chesterfield, VA	0.76	0.88	31
Indian River, FL	0.74	1.23	28
Bay, FL	0.73	0.72	31
Palm Beach, FL	0.73	0.90	25
Pima, AZ	0.72	0.94	30
Charlotte, FL	0.71	0.92	26
Putnam, FL	0.69	0.80	24
Columbia, FL	0.69	1.01	26
Ottawa, MI	0.67	0.88	33
Macomb, MI	0.66	0.98	33
Allegan, MI	0.66	0.94	27
Seminole, FL	0.65	0.47	24
Richmond, GA	0.65	0.28	14
Lee, FL	0.65	0.60	28
Alachua, FL	0.65	0.66	37
Martin, FL	0.65	0.69	29
Leon, FL	0.64	0.65	29
Hall, GA	0.64	0.88	18
Loudoun, VA	0.63	1.31	29
Walton, FL	0.63	0.87	27
Wright, MN	0.62	0.69	11
Santa Rosa, FL	0.62	0.83	19
St. Johns, FL	0.61	0.64	33
Flagler, FL	0.61	0.31	18
Yavapai, AZ	0.61	1.29	29
Osceola, FL	0.61	0.53	21
Blue Earth, MN	0.61	0.54	11

Table A1: Comparison-county weights by outcome, average over 45 assessments.

Notes. Each county's weight averaged over the 45 assessments, in percent, for foreclosure filings and completed foreclosures; a county outside an assessment's pool counts as zero there. The last column counts the assessments in which the county carries filing weight. Counties are sorted by mean filing weight.

	Mean weight over 45 assessments (%)		Assessments
County	Filing	Completed foreclosure	with filing weight
Scott, MN	0.60	1.17	19
Lake, FL	0.60	0.59	27
Livingston, MI	0.59	1.02	22
Warren, KY	0.58	0.63	11
Barry, MI	0.58	1.22	21
Polk, FL	0.57	0.62	28
Washtenaw, MI	0.56	0.82	19
Nassau, FL	0.56	0.35	22
Manatee, FL	0.56	0.94	25
Duval, FL	0.52	0.97	22
Jessamine, KY	0.52	0.73	7
Campbell, KY	0.51	0.38	13
Citrus, FL	0.50	0.60	26
Okeechobee, FL	0.49	1.22	14
Cass, MI	0.49	0.67	13
Maricopa, AZ	0.49	1.19	27
Pinal, AZ	0.48	0.86	19
Hampden, MA	0.48	0.43	15
Sherburne, MN	0.47	0.74	14
Escambia, FL	0.46	0.75	23
Carver, MN	0.45	0.73	17
Plymouth, MA	0.43	0.28	14
Collier, FL	0.42	0.57	24
Mecosta, MI	0.41	0.71	9
Houston, AL	0.40	0.11	8
Suffolk, MA	0.40	0.27	16
Ramsey, MN	0.40	0.52	20
Becker, MN	0.40	0.31	10
Norfolk, MA	0.40	0.36	15
Okaloosa, FL	0.39	0.98	17
Washington, MN	0.39	1.16	17
Dakota, MN	0.38	0.86	17
Middlesex, MA	0.38	0.35	17
Midland, MI	0.37	0.42	11
Monroe, MI	0.34	0.23	13
Worcester, MA	0.34	0.24	13
Essex, MA	0.34	0.20	11
Bristol, MA	0.32	0.14	11
Jackson, GA	0.29	0.30	8
Genesee, MI	0.21	0.00	2

Table A1: Comparison-county weights by outcome, average over 45 assessments.

Notes. Each county's weight averaged over the 45 assessments, in percent, for foreclosure filings and completed foreclosures; a county outside an assessment's pool counts as zero there. The last column counts the assessments in which the county carries filing weight. Counties are sorted by mean filing weight.

	Mean weight over 45 assessments (%)		Assessments
County	Filing	Completed foreclosure	with filing weight
Clay, MN	0.19	0.30	8
Rice, MN	0.18	0.04	2
Stearns, MN	0.16	0.16	5
Anoka, MN	0.15	0.01	2
Shelby, KY	0.14	0.14	2
Lenawee, MI	0.09	0.43	2
Suwannee, FL	0.03	0.21	1

A.2 tax liabilities between assessments

Table A2: Average property tax rates by year, 34 North Carolina counties, 2009–2020.

Year	County	Municipal	County plus municipal	Assessments studied
2009	0.612	0.365	0.947	0.0
2010	0.614	0.366	0.960	0.0
2011	0.623	0.371	0.969	8.0
2012	0.630	0.379	0.982	6.0
2013	0.639	0.384	1.006	3.0
2014	0.646	0.391	1.021	1.0
2015	0.659	0.396	1.038	6.0
2016	0.659	0.401	1.042	6.0
2017	0.657	0.406	1.048	13.0
2018	0.657	0.405	1.049	2.0
2019	0.658	0.408	1.057	0.0
2020	0.654	0.408	1.049	0.0

Notes. Rates per $100 of assessed value in the 34 counties of the 45 assessments. County is the simple average of county rates; municipal and county plus municipal average over the 238 to 253 municipalities in those counties. Special-district rates are not included. Assessments studied counts the assessments in this paper that took effect that year.Between assessments the rate moves by less than three cents a year and no quartile's tax changes by more than 4 percent.

A.3 Appreciation since the previous assessment

Table A3: Appreciation between the previous assessment and this one, cheaper and more expensive halves of homes.

		Appreciation since last assessment			Lower half
Cycle	Events	Lower half	Upper half	Gap (points)	appreciated more
4 years	21	3%	2%	+3	13 (62%)
5 years	5	-5%	-2%	+2	3 (60%)
6–7 years	4	12%	-6%	+9	3 (75%)
8 years	12	7%	4%	+1	7 (58%)
All	42	4%	1%	+3	26 (62%)

Notes. Arm's-length sales of owner-occupied homes. For each value quartile, appreciation is the change in the median log sale price from the county's previous assessment year to the year before the assessment, in percent; a half is the owner-occupied-weighted mean of its two quartiles. Quartiles are each assessment's block-group quartiles. The last column counts the assessments in which the lower half appreciated more. Assessments without sales in the previous assessment year are omitted.

A.4 What follows a foreclosure filing

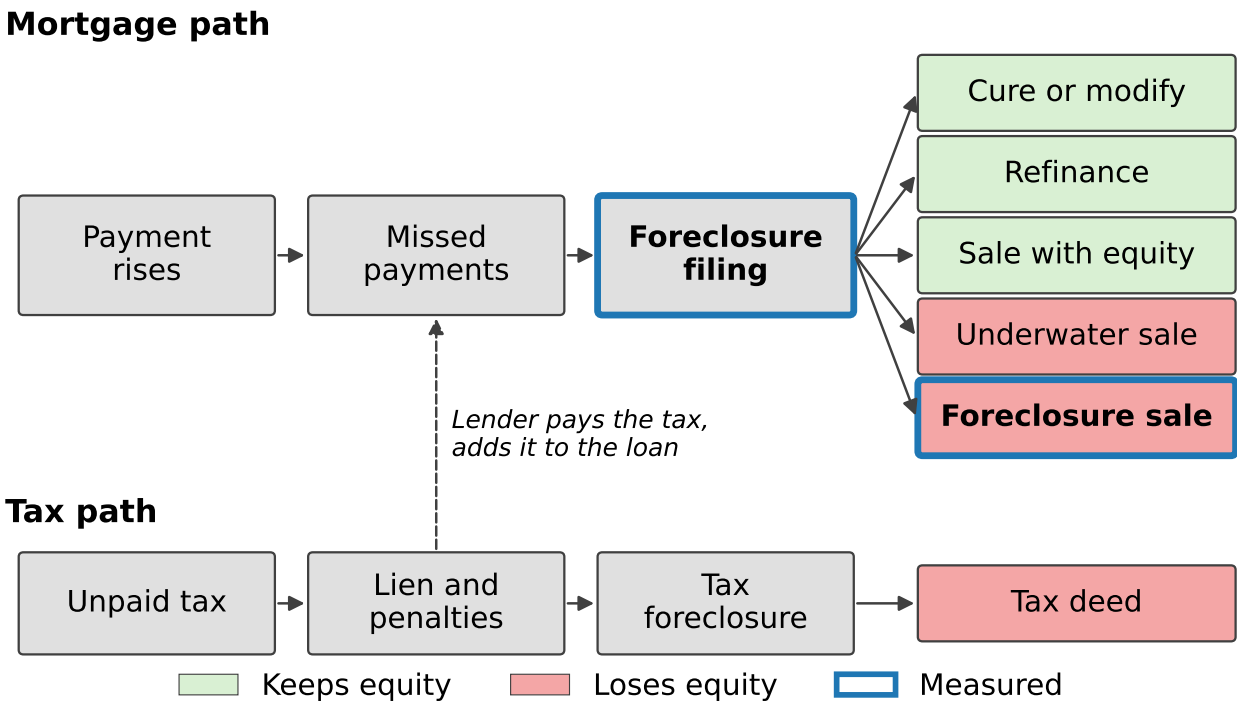


Figure A2: Paths from a missed payment to a recorded outcome.

Notes. Schematic, not estimated. On the mortgage path, a foreclosure filing opens an episode that ends in one of five ways: a cure or loan modification, which leaves no deed on record; a refinance; a sale with equity; an underwater sale; or a foreclosure sale. Green outcomes keep the owner's equity and red ones lose it; outlined boxes are the outcomes this paper measures. On the tax path, which applies to homes with no mortgage or no escrow account, unpaid property tax leads to a county lien and a tax deed without a foreclosure filing; where a lender holds the loan, it usually pays the tax and adds it to the balance, which moves the case to the mortgage path.

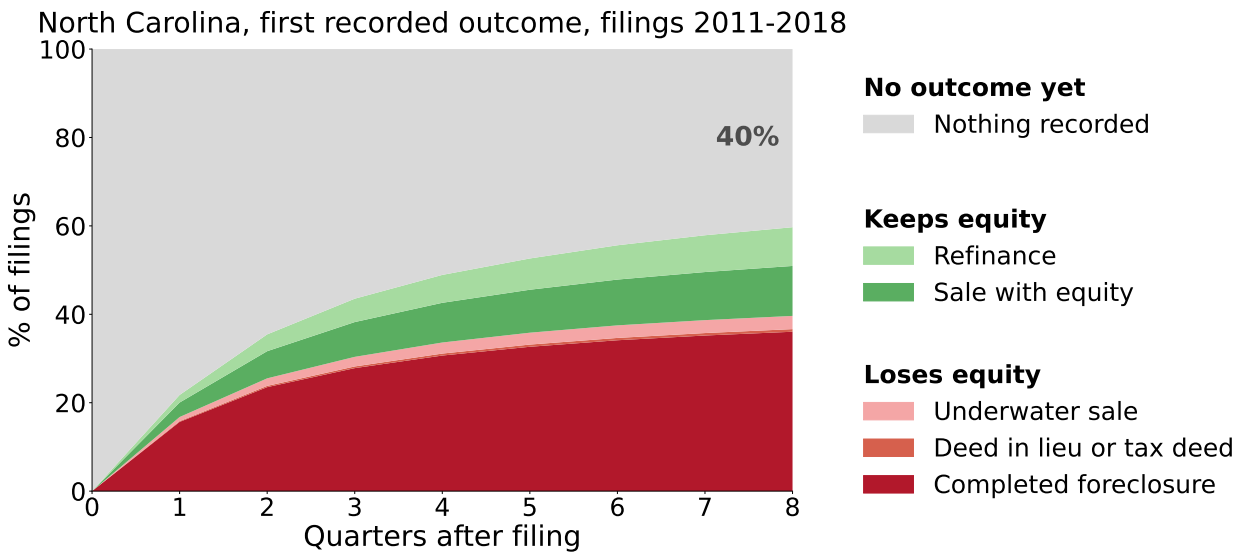


Figure A3: First recorded outcome after a foreclosure filing, North Carolina.

Notes. Cumulative share of filing episodes on owner-occupied homes whose first later document is each outcome, by quarter after the filing, pooling Mecklenburg, Guilford, Forsyth, Durham and Wake and all four value quartiles. Filings from 2011 to 2018, followed through 2020. An episode is a filing with no filing on the same parcel in the preceding eight quarters. An underwater sale is an arm's-length sale below the estimated payoff, the mortgage debt plus arrears and selling costs; a refinance is a new mortgage with no change of title before it, and includes recorded modifications. The unfilled share has no recorded outcome by that quarter.

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